

Safety Timer for Home Appliances

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When power requirement in certain areas increases, it tends to put an extra burden on the power station distributing power, which can lead to over and under voltage issues within those areas. This situation of over and under voltages is not good for home appliances such as TV, refrigerator, computer and water motor as all of them should run only within a safe range of input voltage to ensure their long life and functionality.

This project not only ensures under and over voltage cutoff, but also displays the mains voltage over LCD. The device is equipped with a delay timer which allows it to re-start only after a period of time elapses. This ensures safety for conditions when the



voltage drops or rises very rapidly and can harm sensitive appliances. Re-start time is selectable from 30 seconds to a maximum of 4 minutes.

The under and over voltage cut-off limits are programmable and the user can change them according to the load voltage endurance. When the mains input voltage leaves the safe operating area, the output load is immediately turned off and the user is warned through audio-visual indication. The device works satisfactorily from 100V to 300V. The unit is also equipped with a temperature sensor that senses the room temperature, which is displayed

on the LCD.

Fig. 1 shows the author's prototype and Fig. 2 shows the overall block diagram.

Circuit and working

Fig. 3 shows circuit diagram of the safety timer for home appliances. It is built around microcontroller ATmega8 (IC1), LM35 temperature sensor (IC2), 15V, 500mA and 12V, 1A step-down transformers (X1 and X2, respectively), 5V voltage regulator 7805 (IC3) and LCD.

Transformer X2 is used to power the circuit. The secondary of X2 is connected to BR2, which is a general-purpose 1A bridge rectifier. The DC output is filtered through capacitor C1 and fed to 5V regulator IC3 to get regulated 5V rail to power all the components including microcontroller.

For stability of IC3, electrolytic capacitor C2 and ceramic capacitor C3 are used. LED1 is used to indicate the input power status of the circuit; current through it is limited by resistor R1.

Transformer X1 is used to sense the mains input voltage. The secondary output is fed to bridge rectifier BR1 and the DC value is then scaled down using the voltage divider network formed by resistor R3 and pre-set VR1. The divided voltage is

PARTS LIST

Semiconductors:

- IC1 - ATmega8 microcontroller
- IC2 - LM35 temperature sensor
- IC3 - 7805, 5V regulator
- LED1, LED2 - 5mm LED
- D1 - 1N4001 rectifier diode
- ZD1 - 5V Zener diode
- T1 - BC547 npn transistor
- T2 - BC558 pnp transistor
- LCD1 - 16x2 LCD
- BR1, BR2 - Bridge rectifier module, 1A

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 680-ohm
- R2 - 330-ohm
- R3 - 100-kilo-ohm
- R4 - 10-kilo-ohm
- R5 - 2.2-kilo-ohm
- R6 - 4.7-kilo-ohm
- VR1 - 15-kilo-ohm preset
- VR2 - 10-kilo-ohm preset

Capacitors:

- C1 - 2200 μ F, 35V electrolytic
- C2, C6 - 10 μ F, 16V electrolytic
- C3, C5 - 0.1 μ F ceramic disk
- C4 - 0.47 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 15V, 500mA secondary transformer
- X2 - 230V AC primary to 12V, 1A secondary transformer
- S1-S5 - Tactile switch
- CON1, CON2 - 2-pin 5mm terminal connector
- PZ1 - Piezo buzzer
- RL1 - 5V, 1 C/O relay



Fig. 1: Screenshot of the author's prototype

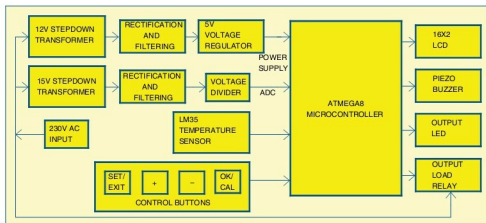


Fig. 2: Block diagram of safety timer for home appliances